ABCD Purity Bootstrap — Uncertainty Quantification

Comparison of Gaussian fit, Crystal Ball, and equal-tail percentile (median + 16/84%)

PPG12

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Question

The current PPG12 ABCD purity is reported as the mean and width of a Gaussian fit to a 20 000-throw Poisson bootstrap of the four region yields (A, B, C, D) plus Gaussian smearing of the signal-leakage corrections (c_B, c_C, c_D). The bootstrap distribution is bounded above by $P \leq 1$, and at high E_T^γ the truncation produces a long lower tail that the symmetric Gaussian fit does not represent faithfully. This note compares three uncertainty-quantification choices, all driven by the *same* bootstrap sample, and recommends the equal-tail percentile (median + 16/84%) as a shape-agnostic alternative.

Methods

All three methods consume the same per-bin bootstrap histogram `h_purity_leak_`i (or `h_purity_`i for the no-leakage variant) stored in `Photon_final_bdt_nom.root`. Code path: `efficiencytool/CalculatePhoto` lines 430–654; nominal closed-form values reproduced in `plotting/plot_purity_bootstrap.C`.

1. **Gaussian fit (current default).** `TF1("gaus")` fitted in $[\mu - 1\,\mathrm{RMS},\ \mu + 1.5\,\mathrm{RMS}]$. Central value μ, error σ. Symmetric. Currently written to `gpurity_leak` and propagated downstream.
2. **Crystal Ball (single-sided low tail).** On-the-fly fit with parameters $(N, \mu, \sigma, \alpha, n)$, range $[\mu - 8\,\mathrm{RMS},\ \mu + 1.5\,\mathrm{RMS}]$, likelihood fitter. Captures the asymmetric lower tail; reduces to Gaussian when $\alpha \to$large (low-E_T^γ regime). Asymmetric in shape but reported as a single σ.
3. **Equal-tail percentile (median + 16/84%).** Non-parametric. Mid-point is the 50% quantile of the bootstrap; uncertainty is the asymmetric pair $(P_{50} - P_{16},\ P_{84} - P_{50})$. No fit, no shape assumption. Trivially captures truncation against a hard boundary.

The closed-form (no-toy) nominal P_{nom} from solving $N_A^{\mathrm{sig}} = N_A - R\,(N_B - c_B N_A^{\mathrm{sig}})\,(N_C - c_C N_A^{\mathrm{sig}})/(N_D - c_D N_A^{\mathrm{sig}})$ at central yields ($R = 1$ as in the macro) is shown alongside as a green dashed line in every panel for reference.

Per-bin comparison (leakage-corrected)

Table 1: ABCD purity per E_T^γ bin (with $c_B/c_C/c_D$ smearing). All methods consume the same 20 000-throw bootstrap. Gaussian columns are the fit $\mu \pm \sigma$. Median columns report P_{50} with the asymmetric 1σ-equivalent percentile pair $\binom{+(P_{84}-P_{50})}{-(P_{50}-P_{16})}$. P_{nom} is the closed-form nominal.

bin	E_T^γ (GeV)	P_{nom}	μ_{Gaus}	σ_{Gaus}	P_{50}	$-1\sigma_{\text{pct}}$	$+1\sigma_{\text{pct}}$
1	11	0.5098	0.5099	0.0135	0.5081	0.0138	0.0134
2	13	0.5613	0.5618	0.0174	0.5598	0.0180	0.0172
3	15	0.6207	0.6216	0.0223	0.6192	0.0235	0.0223
4	17	0.6963	0.6986	0.0284	0.6950	0.0307	0.0285
5	19	0.6732	0.6772	0.0430	0.6725	0.0467	0.0413
6	21	0.6197	0.6291	0.0798	0.6193	0.0913	0.0766
7	23	0.8188	0.8276	0.0552	0.8187	0.0688	0.0526
8	25	0.8499	0.8634	0.0656	0.8520	0.0920	0.0605
9	27	0.8633	0.8714	0.0626	0.8692	0.1253	0.0720
10	30	0.7170	0.7344	0.1419	0.7293	0.3062	0.1592
11	34	0.9730	0.9848	0.0727	0.9872	0.1553	0.0511

Reading the table.

- **Bins 1–5** (E_T^γ 10–20 GeV): all three methods agree to within 0.001 in central value, the percentile band is symmetric to within $\sim 5\%$. The bootstrap is genuinely Gaussian — no method is preferable on shape grounds.
- **Bins 6–9** (20–28 GeV): asymmetry sets in. The -1σ percentile width exceeds the $+1\sigma$ by 12–74%. Gaussian σ underestimates the lower bound and overestimates the upper bound for these bins.
- **Bin 10** (28–32 GeV): strongly asymmetric. $-1\sigma_{\text{pct}} = 0.306$ vs $+1\sigma_{\text{pct}} = 0.159$ (factor-2 asymmetry). The Gaussian $\sigma = 0.142$ averages the two and misrepresents both.
- **Bin 11** (32–36 GeV): pathological. The bootstrap is bunched against $P = 1$, so $P_{84} = 1.038 > 1$ (i.e., 16% of toys give a leakage-corrected purity above 1, mathematically allowed by the Gaussian smearing of $c_B/c_C/c_D$ but unphysical). This is itself a useful diagnostic that the Gaussian fit hides.

Figures

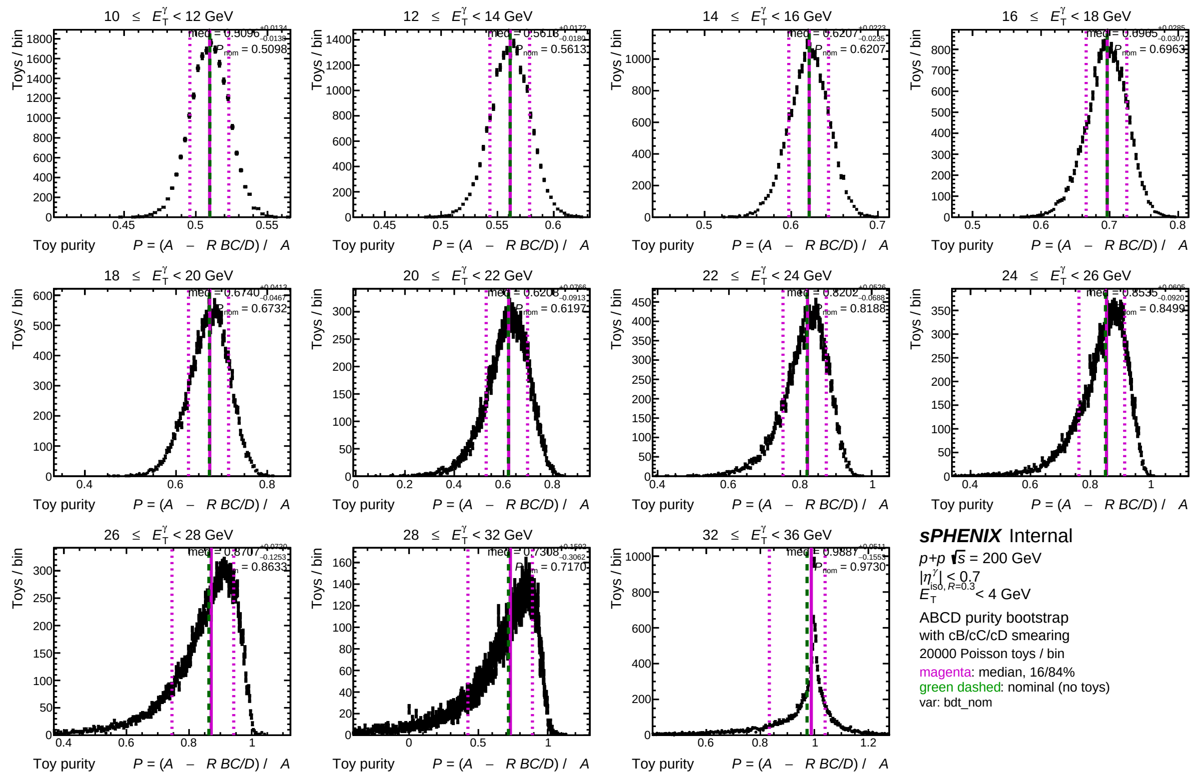

Figure 1: Bootstrap distributions per E_T^γ bin (leakage-corrected), overlaid with the equal-tail percentile band: solid magenta line at P_{50}, dashed at P_{16} and P_{84}. Green dashed line is the closed-form nominal $P_{\rm nom}$. Asymmetry between $P_{50}-P_{16}$ and $P_{84}-P_{50}$ is visible from $E_T^\gamma \geq 22\,\mathrm{GeV}$ onward; in the top bin the upper percentile crosses $P=1$.

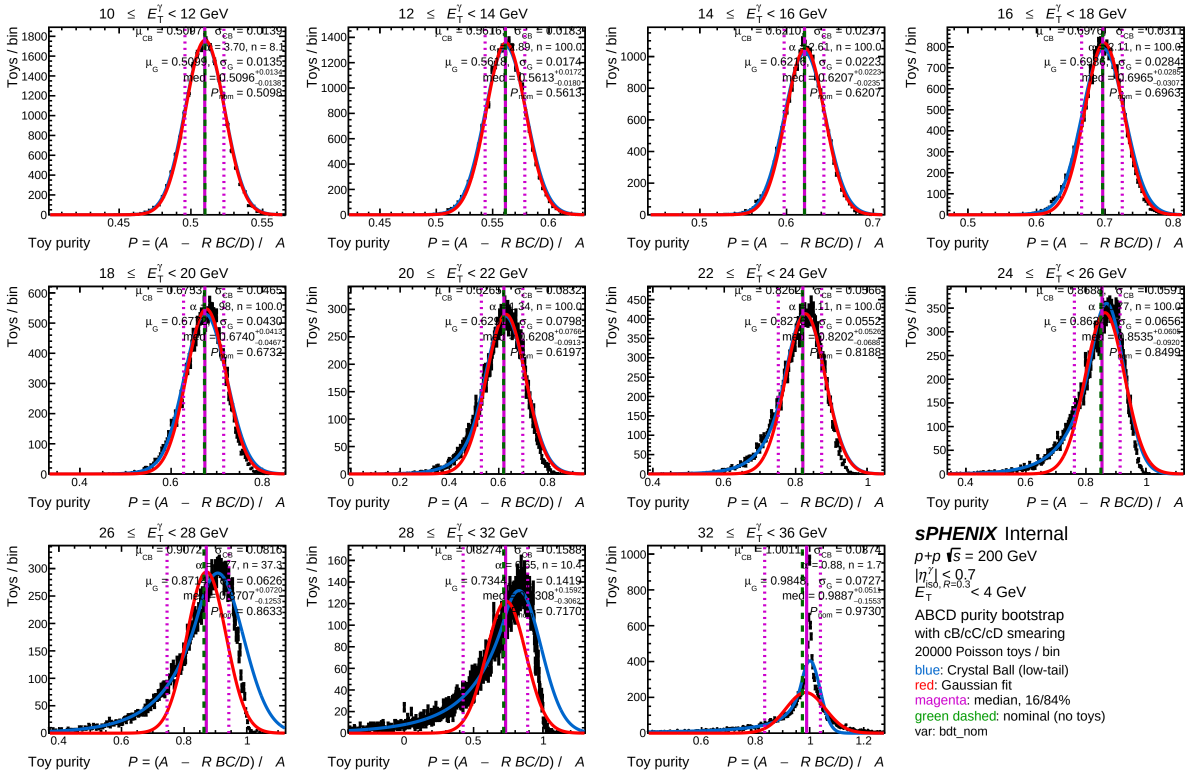

Figure 2: Same data, all three uncertainty estimators overlaid. Red: Gaussian fit $\mu \pm \sigma$. Blue: Crystal Ball low-tail fit. Magenta: median + 16/84% band. Green dashed: closed-form nominal. The three methods are visually indistinguishable for $E_T^\gamma < 20$ GeV; CB and percentile diverge from Gaussian above 22 GeV.

Recommendation

Switch the published σ_P in `gpurity_leak` to the equal-tail percentile, with median as the central value, for the final cross-section. Three reasons:

1. Shape-agnostic — no Gaussian assumption that demonstrably fails at high E_T^γ.
2. Asymmetric error bars are physically meaningful when the underlying distribution is asymmetric. The downstream cross-section Padé fit currently uses symmetric σ; switching to the asymmetric percentile pair propagates the truncation effect cleanly rather than absorbing it into a fictitious symmetric width.
3. Robust against the boundary effect: $P_{84} > 1$ in the top bin is reported transparently rather than implicitly clipped by a Gaussian fit that doesn't know about the bound.

Caveats:

- For low-E_T^γ bins where the bootstrap is symmetric and Gaussian, the percentile method gives the same answer to within Monte Carlo noise. There is no penalty for using percentile everywhere.
- Median is always inside the percentile interval (true for any unimodal distribution). The pair $(P_{50}, [P_{16}, P_{84}])$ is the "percentile credible interval"; it is not the HPD interval but is more commonly used in HEP and is invariant under monotonic transformations.
- The current downstream Padé fit (`f_purity_leak_fit`) assumes symmetric per-point errors. Migrating to the asymmetric percentile requires either symmetrizing via $\sigma_{\text{eff}} = \frac{1}{2}(P_{84} - P_{16})$ for the fit and reporting the asymmetric errors only on the per-bin points, or using `TGraphAsymmErrors` with a fitter that supports it.

Code and reproducibility

- Plotting macro: plotting/plot_purity_bootstrap.C (modes: gaus, cb, pct, both, all). Default mode is cb. To regenerate the percentile figure:

```
cd plotting && root -l -b -q \
  'plot_purity_bootstrap.C("bdt_nom","data_histo","pct")'
```

- Output figures live in plotting/figures/.
- Numerical re-derivation of the percentile values in this note was performed via uproot + numpy (linear- interpolated quantile of the cumulative distribution, matching TH1::GetQuantiles).
- Bootstrap RNG seed is fixed at 42 (CalculatePhotonYield.C:430), so all numbers in Tab. 1 are reproducible bit-for-bit by re-running the analysis pipeline on the same input.